

# Practice Questions for 04L1-Linear Algebra Primer

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## 1 Questions

1. According to the lecture, what is the dimension of a matrix with  $n$  columns and  $m$  rows?
  - A.  $n \times m$
  - B.  $m \times n$
  - C.  $n + m$
  - D.  $n - m$
  - E.  $m/n$
2. If a linear transformation  $T$  is defined as  $T(x) = Ax$ , where  $A$  is a matrix, which of the following properties must it satisfy?
  - A.  $T(u + v) = T(u) + T(v)$  and  $T(cu) = cT(u)$
  - B.  $T(u + v) = T(u) - T(v)$  and  $T(cu) = cT(u)$
  - C.  $T(u + v) = T(u)T(v)$  and  $T(cu) = c + T(u)$
  - D.  $T(u + v) = T(u) + T(v)$  and  $T(cu) = c + T(u)$
  - E.  $T(u + v) = T(u) - T(v)$  and  $T(cu) = c - T(u)$
3. What is the result of multiplying a matrix  $A$  of dimensions  $m \times n$  with a matrix  $B$  of dimensions  $n \times p$ ?
  - A. A matrix of dimensions  $n \times m$
  - B. A matrix of dimensions  $m \times p$
  - C. A matrix of dimensions  $p \times m$
  - D. A matrix of dimensions  $n \times p$
  - E. The matrices cannot be multiplied

## 2 Answers

1. According to the lecture, what is the dimension of a matrix with  $n$  columns and  $m$  rows?
  - A. The dimensions are given as rows by columns, so this is incorrect.
  - B. The lecture states that a matrix with  $n$  columns and  $m$  rows has dimensions  $m \times n$ .
  - C. The dimensions are not the sum of the rows and columns.
  - D. The dimensions are not the difference of the rows and columns.
  - E. The dimensions are not the ratio of the rows and columns.
2. If a linear transformation  $T$  is defined as  $T(x) = Ax$ , where  $A$  is a matrix, which of the following properties must it satisfy?
  - A. The lecture states that a linear transformation  $T$  must satisfy  $T(u+v) = T(u)+T(v)$  and  $T(cu) = cT(u)$  for all vectors  $u, v$  and scalars  $c$ .
  - B. The first property is incorrect, it should be addition not subtraction.
  - C. Both properties are incorrect.
  - D. The second property is incorrect, it should be multiplication not addition.
  - E. Both properties are incorrect.
3. What is the result of multiplying a matrix  $A$  of dimensions  $m \times n$  with a matrix  $B$  of dimensions  $n \times p$ ?
  - A. The dimensions are incorrect.
  - B. The lecture states that the product of an  $m \times n$  matrix  $A$  and an  $n \times p$  matrix  $B$  is an  $m \times p$  matrix.
  - C. The dimensions are incorrect.
  - D. The dimensions are incorrect.
  - E. The matrices can be multiplied because the inner dimensions match.